

```

}

static void __exit mod_par_exit(void)
{
    printk(KERN_INFO "Unloaded cfg value: %d\n", cfg_value);
}

module_init(mod_par_init);
module_exit(mod_par_exit);
MODULE_LICENSE("GPL");
MODULE_AUTHOR("Anil Kumar Pugalia <email@sarika-pugs.com>");
MODULE_DESCRIPTION("Module Parameter demonstration Driver");

```

Note that before the parameter set-up, a variable of the same name and compatible type needs to be defined. Subsequently, the following steps and experiments are shown in Figures 2 and 3:

- Building the driver (*module_param.ko* file) using the usual driver *Makefile*
- Loading the driver using *insmod* (with and without parameters)
- Various experiments through the corresponding */sys* entries
- And finally, unloading the driver using *rmmod*.

Note the following:

- Initial value (3) of *cfg_value* becomes its default value when *insmod* is done without any parameters.

- Permission *0764* gives *rxw* to the user, *rw-* to the group, and *r--* for the others on the file *cfg_value* under the parameters of *module_param* under */sys/module/*.

Check for yourself:

- The output of *dmesg / tail* on every *insmod* and *rmmod*, for the *printk* outputs.
- Try writing into the */sys/module/module_param/parameters/cfg_value* file as a normal (non-root) user.

Summing up

With this, the duo have a fairly good understanding of Linux drivers, and are all set to start working on their final semester project. Any guesses what their project is about? Hint: They have picked up one of the most daunting Linux driver topics. Let us see how they fare with it. **END** 

By: Anil Kumar Pugalia

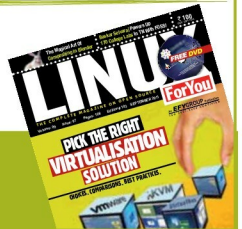
The author is a freelance trainer in Linux internals, Linux device drivers, embedded Linux and related topics. Prior to this, he was at Intel and Nvidia. He has been working with Linux since 1994. A gold medallist from the Indian Institute of Science, Linux and knowledge-sharing are two of his many passions. Creating and playing with open source hardware is one of his hobbies. Learn more about him at profession.sarika-pugs.com/, including information about his open source hardware freak-outs and his various Linux-related training sessions and workshops. He can be reached at email@sarika-pugs.com.

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